

May 2024

Powerledger Solutions for Transactive Energy: P2P Energy Trading

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strictly confidential

Agenda

- SmartCommunity
- Vision
- Web3



Major Takeaway



- Powerledger has a unique proposition and regulations are no hurdles to implement.
- [SmartCommunity](#) solution will help your business clients to win more, and you'll retain more. It also allows you to do that uniquely by keeping prices low: hedging against price volatility, empower them through P2P between them and help them with end to end carbon reporting using [Vision](#).
- Take these steps now and future proof by becoming [Smart-Web3](#) enabled retailer.

[Powerledger](#)

SmartCommunity

Distributed Energy Exchange Solution



SmartCommunity: objectives

- Enable green energy adoption for residential and business customers via SmartCommunity.
- Provide transparency on energy sourcing and destination.
- Democratiser renewable energy production by involving end-users.
- Offer risk management tools for businesses against market price volatility.

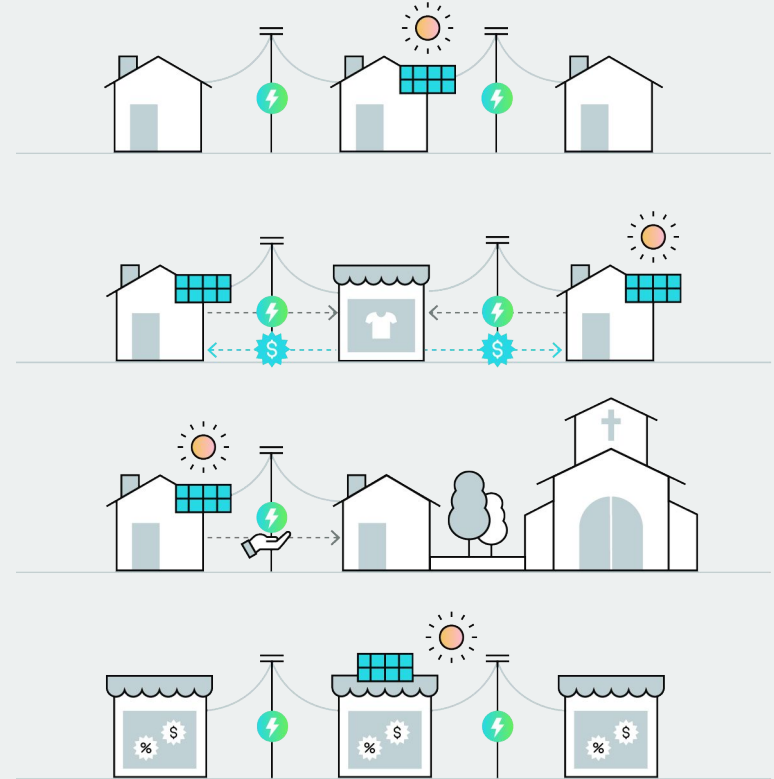
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SmartCommunity - Key Benefits

- Business clients get more cheaper renewable
- Our Customer will retain more customers
- Uniquely keep the prices low: hedging against price volatility
- Empower customers to do P2P between them through you

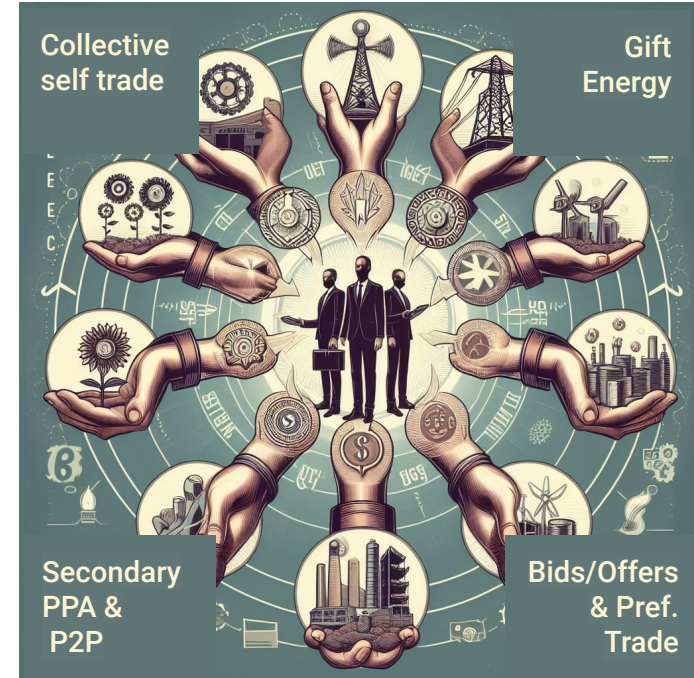
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SmartCommunity - Market mechanisms

- Collective Self-Trade
- Secondary Power Purchase Agreements
- Peer to peer
- Control over pricing - bids and offers
- Control over buyer/seller - preferential trades
- Gifting energy



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ESTMK & Powerledger partnership - SmartCommunities in Austria

- Monitor usage, consumption, generation, transaction profile, and their savings/revenue.
- Create customized/tailored Power Purchase Agreements (PPAs) with other members in their energy community
- Preferentially consume their own generation against their consumption, before trading with others, and finally with the grid.
- (In development) Support local businesses, causes etc. by preferentially trading with said businesses/organizations through the Powerledger Loyalty feature.



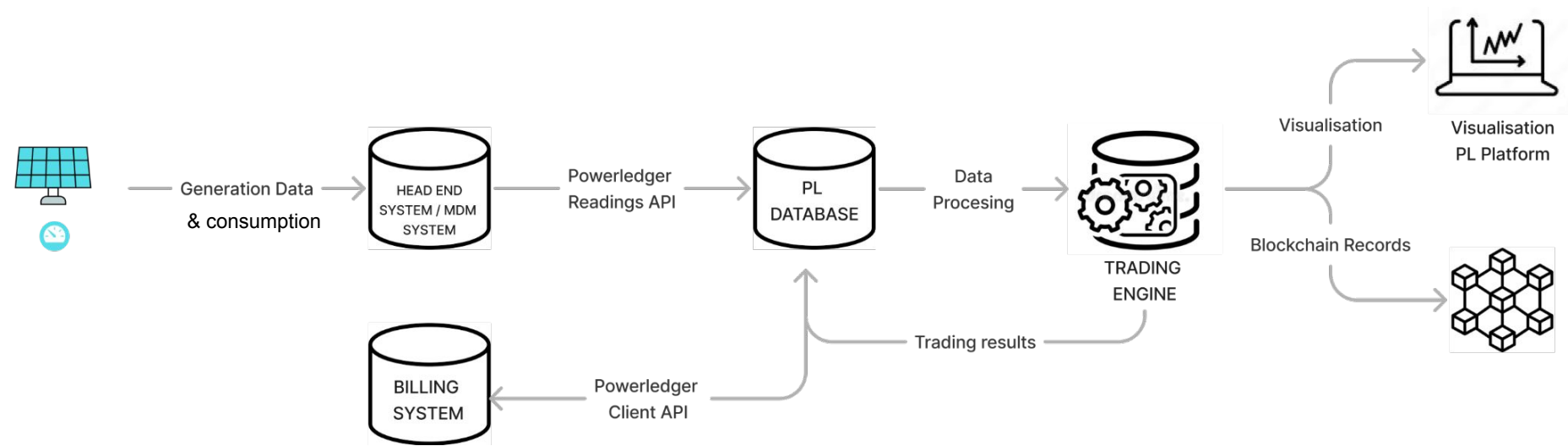
Smart Communities Austria: Savings

- Average savings for **B2C** customers: €140/ MWh P2P
- Average savings for **B2B** customers: €230/ MWh P2P
- **B2C** contracts typically have **fixed tariffs** and highest savings per meter per month are ~ €6
- **B2B** contracts with **variable pricing** making them more exposed to the market, highest savings per meter per month are ~ €1100
- Savings depend on pricing of p2p relative to grid tariffs and available volume of p2p energy

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Platform Data Flow for demonstration project



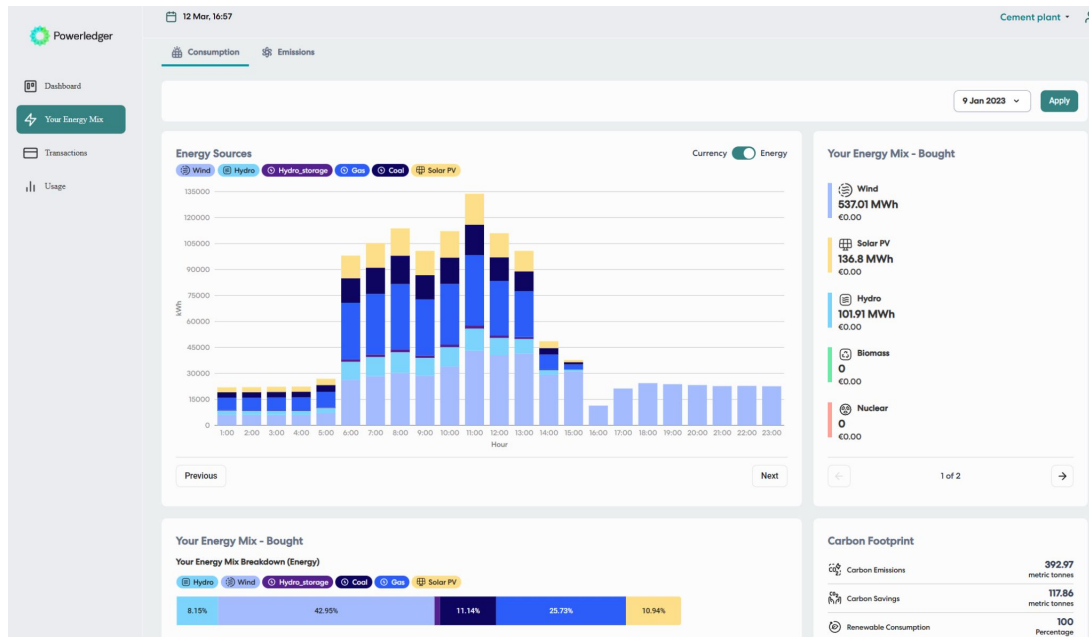
Vision

Powerledger's Decarbonisation Solution



What is Vision

- Automated visualisation tool for energy consumption and carbon emissions reporting.
- Provides breakdown of energy types by location and time interval at meter level.
- Matches real-time energy consumption with available renewable sources.
- Calculates carbon intensity and assesses Carbon Free Energy score, emphasising Scope 2 emissions.



Screenshot of Vision user interface of platform from pilot. The dashboard visualises the energy consumed for an industrial site, providing a breakdown per hour, per energy type.

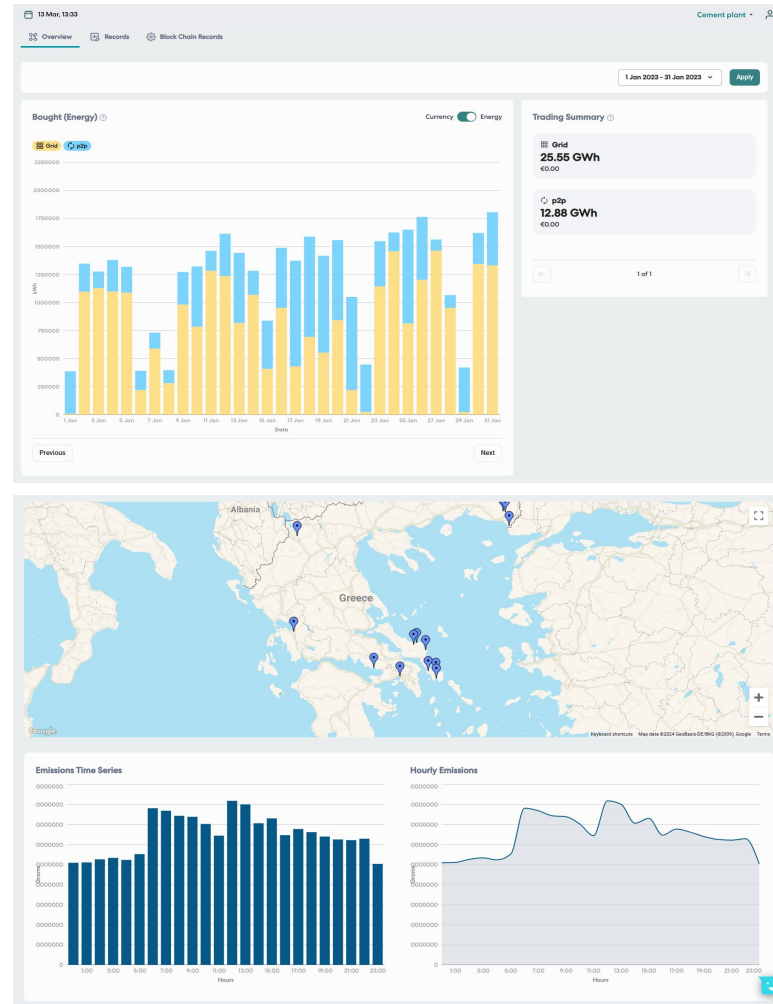
Purpose of Vision

- Vision offers insights on Our Customer's asset synergy and demand fulfillment.
- Integrates grid and emissions data for hourly Scope 2 emissions estimation, aiding decarbonization.
- Incorporates pricing data for financial insights, informing decision-makers on arrangement performance.

Screenshot of Vision user interface of platform from pilot:

Top image shows how Ekwatuer RE vs grid energy fulfills an industrial client's energy demand.

Bottom image displays the estimated emissions time series



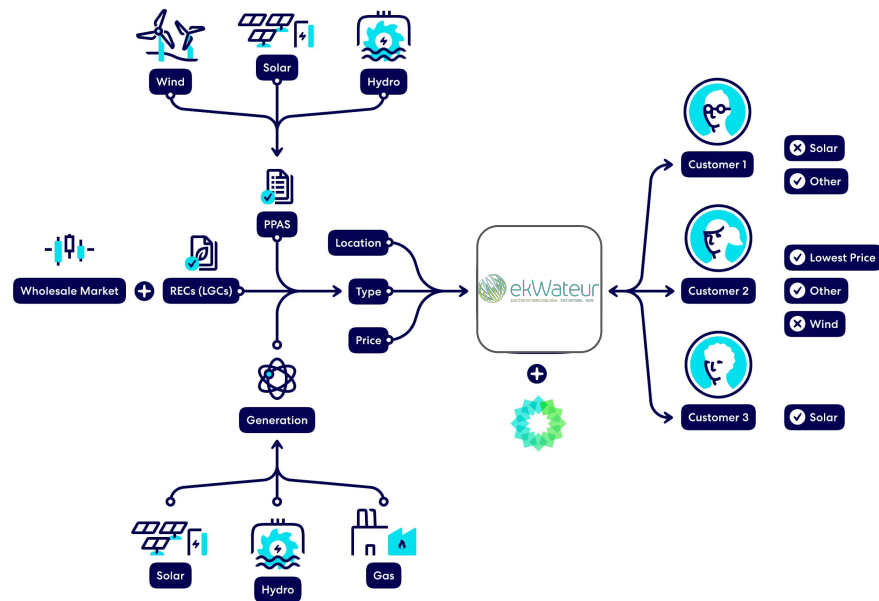
Value of Vision

Effective decision-making relies on data availability and the insights drawn from it. The first step towards 24/7 RE is a robust accounting system, this is what Vision offers.

Blockchain technology enables an energy tracking system to operate more efficiently by using smart contracts to automate complex processes. At the same time, data stored on a blockchain offers an immutable, transparent and secure audit trail, which promotes trust and accountability. True decarbonisation is achieved by advancing towards 24/7 carbon-free energy, where every kilowatt-hour of electricity consumed is accounted for.

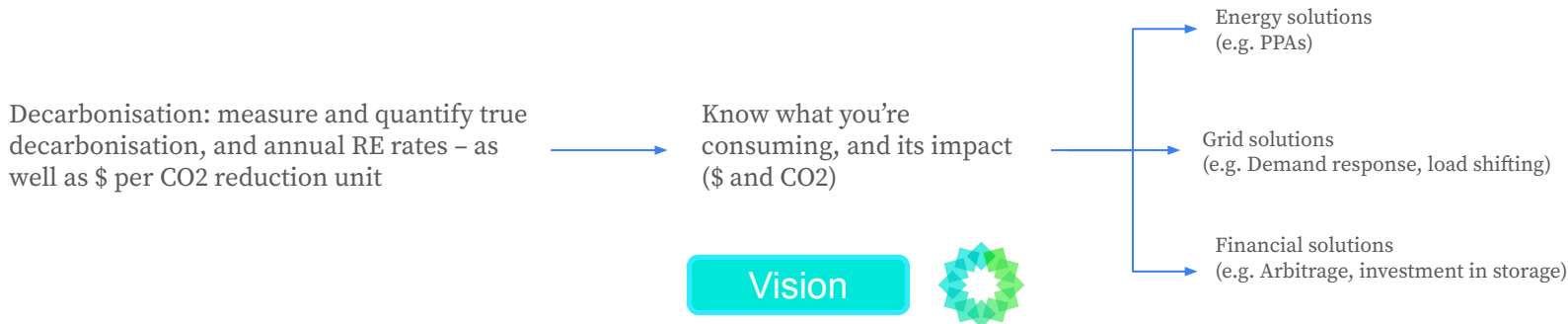
The main benefits Vision are:

- Provides visibility and transparency of energy usage, generation and costs
- Highly configurable energy matching algorithm
- Visualise carbon emissions breakdown and source
- Creates transparency in billing determinants for auditing



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Decarbonisation – with transparent traceability of emissions at its core



An underlying platform to:

- Determine best service for Our Customer's client based on \$ per reduced kgCO2
- Securely track and trace all energy supply
- Hourly and yearly RE matching
- Also allows tokenisation across services/ business units

Vision current implementations



Vision for Blockchain-enabled energy tracking



Shell in partnership with Powerledger has launched Vision as a PoC decarbonisation tool for their Micro-Grids and electrification businesses.

Shell sustainability team can trace and visualise in real-time and report to their commercial customers, where their energy comes from in near real-time, and track their carbon footprint granularly.

With Powerledger's technology, Shell can now effectively offset any consumption for their customers that is non-renewable and create an immutable/auditable trail or records using leveraging the power of blockchain.

Vision for entire Portfolio tracking



MYTILINEOS
Energy & Metals

MYTILINEOS in partnership with Powerledger is introducing first time energy provenance tracking and tracing features in Greece through Protergia, its energy retail subsidiary.

Protergia's consumers can easily trace and visualise where their energy comes from in near real-time, and track their carbon footprint granularly.

With Powerledger's technology, MYTILINEOS can get an onestop view of their and Protergia's clients' entire energy portfolio, including onsite generation, PPAs and GOs, offering an efficient way of meeting energy targets at a low cost.

Vision for campus wide sustainability



A smart city project involving three campuses and 142 buildings, leading to 30% renewable energy autonomy. 13MW of solar, and 1.2 MWh of battery capacity across the campus.

Energy source tracking, matching and P2P trading across the campus buildings and faculties.

Granular carbon emissions intensity tracking and visualisation.

Overall reporting and governance on energy consumption.

Smart-Web3

Powerledger's Blueprint for Our
Customer



Envisioning web3 transformation

Leverage the blockchain/AI tech that underpins everything that Powerledger offers.

Our Customer then becomes:

- the master of their data
- the aggregator of their assets
- the enabler of comprehensive solutions

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Why blockchain and AI?

- Increased transparency
- Breakdown of information silos
- Increase efficiency through task automation
- Fraud protection
- Guaranteed business continuity
- Global leader in innovation
- Enhanced efficiency
- Data driven insights
- Improved overall experience and unmatched brand strength

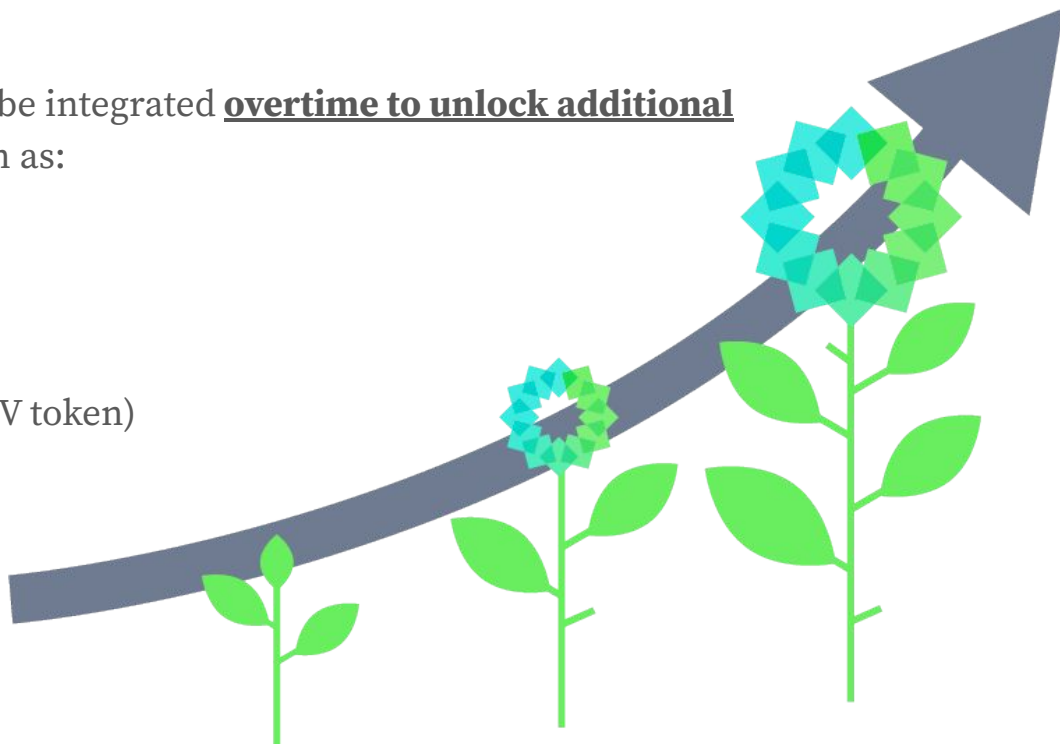
Vision is a gateway towards web3



The first step in your web3 transition consists of storing your data on the Powerledger blockchain.

Multiple optional software products can be integrated **overtime to unlock additional value** for your customers, with tools such as:

- Peer-to-Peer trading
- PPA Optimisation
- Carbon credits exchange
- GO exchange
- Tokenisation-based services (e.g. EV token)



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P2P energy trading options (SmartCommunity - as mentioned before)

Trade energy across the grid, benefitting from local market prices e.g. dynamic pricing (auction pool)/ fixed price / Time of Use tariffs, etc

P2P Anyone with or without solar panels can trade electricity with their peers.

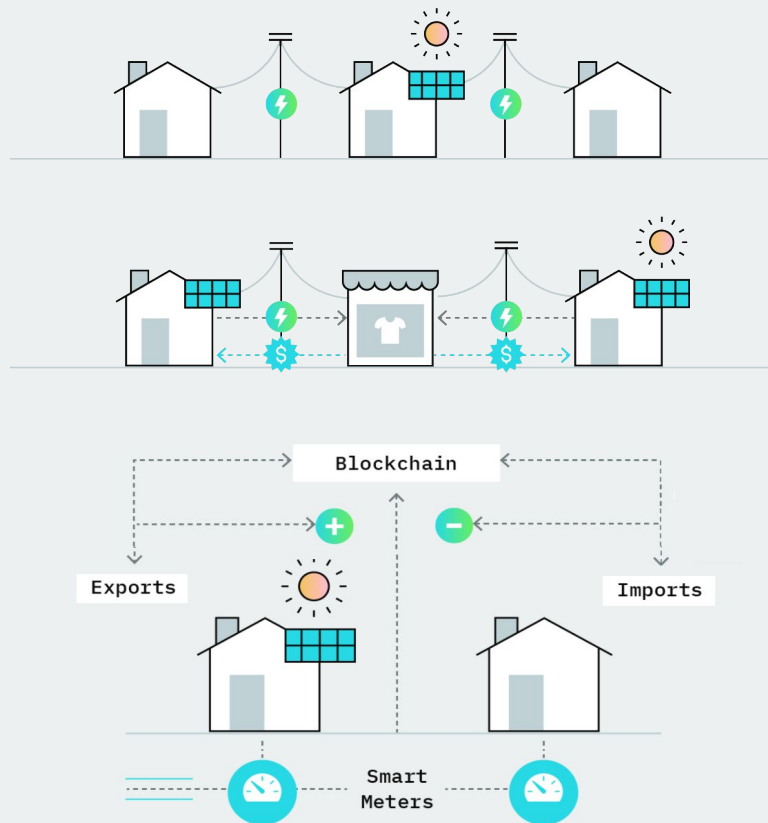
Solar Swap Retail brands can buy excess energy from their clients in exchange for products or vouchers.

Preferential trading Prosumers and consumers can establish their preferred buyers and sellers, even set prices

Gifted P2P Customers can gift electricity to friends, neighbours, schools and community groups.

Cross Store P2P Commercial customers can share solar between their stores located within the same grid, even when only one of them has a PV installation.

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Managing PPA portfolios

Powerledger's Vision (as mentioned before) solution will provide Our Customer with the tools required to obtain a granular tracking of a portfolio of PPA's or renewable production and any further that might be developed in the future:

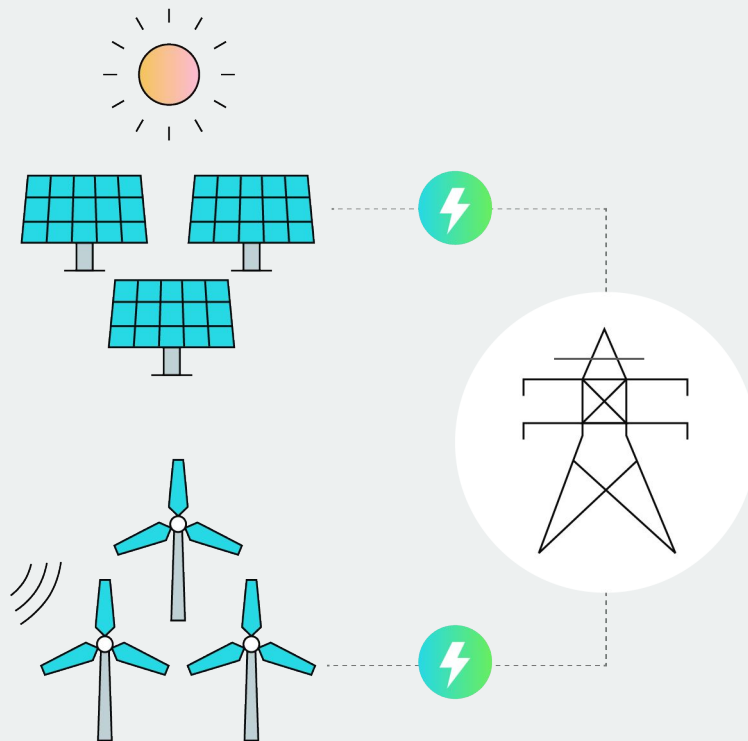
A more streamlined PPA management system with:

- Traceability of generation and offtaking — any size and location
- Removed manual calculation and reconciliation
- One source of truth for mutual benefits to generators and internal departments
- Real-time visibility of billing data
- Reduced costs for audit and compliance

All data is written on blockchain, with immutable records:

- Access to previous period statements and underlying data
- Reporting from the Powerledger Platform or integration to BI systems
- Secured energy certification for audits and carbon accounting

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International Carbon Credits trading - TraceCC

The power utilities environment has become highly international, and TraceCC empowers to play a pivotal role in the international arena through the Carbon Trading space.

Seamless trades

- Place buy and sell orders for Carbon Credits, against those already available in the market dashboard.
- Supports multiple order types and expiry, as well as off-platform trades (OTC settlement).
- Matched orders are seamlessly executed via an in-built clearing and settlement process.

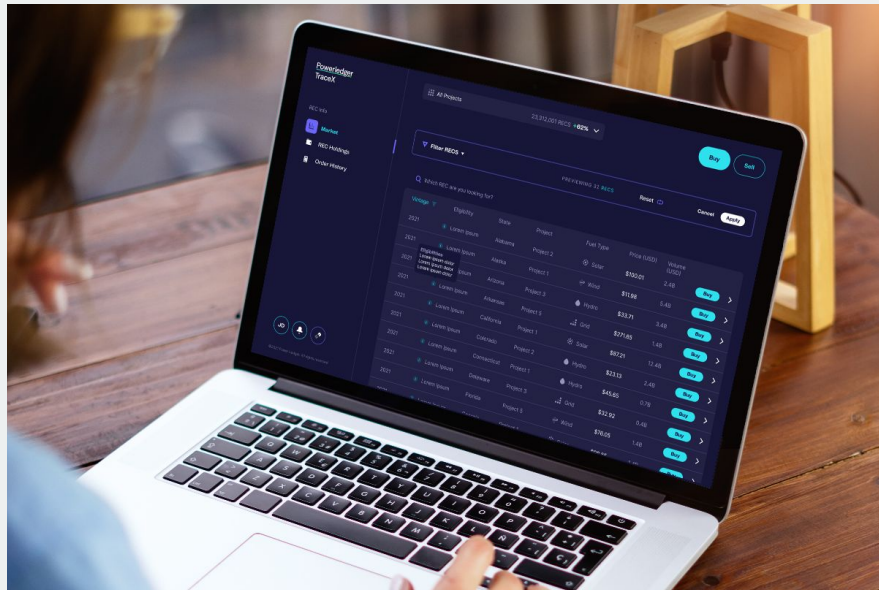
Carbon Credit Holdings

- List your CCs by submitting a simple form in-platform
- Once approved by a marketplace operator, access your CCs from the CC holdings page

Immutable Blockchain records:

- All data is written on the blockchain, ensuring immutable records. TraceCC captures data related to import and export trades, along with retirement events of environmental attributes.

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Voluntary GOs trading - TraceX

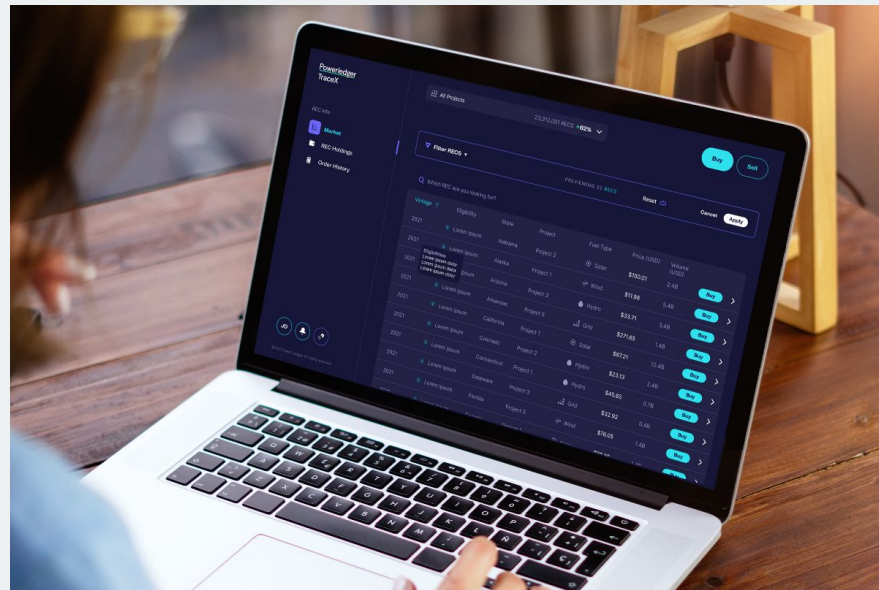


While Guarantees of Origin (GOs) are increasingly key to achieving “net-zero” goals, the current OTC model has limitations.

Traditional GO buying and selling can be:

- **Opaque:** finding buyers and sellers at the right time and the right price is challenging.
- **Expensive:** direct or hidden back-of-house costs.
- **Invasive:** sellers may face credit checks and 3rd party reviews of back-office books.
- **Cumbersome:** high-friction, high-paperwork experience that wastes time and invites error
- **Unreliable:** unpredictable availability of compliant GOs; dependent on both parties being able undertake due diligence activities in a timely manner

TraceX is a blockchain enabled solution that tokenises GOs and bring transparency to the voluntary energy certificate market.



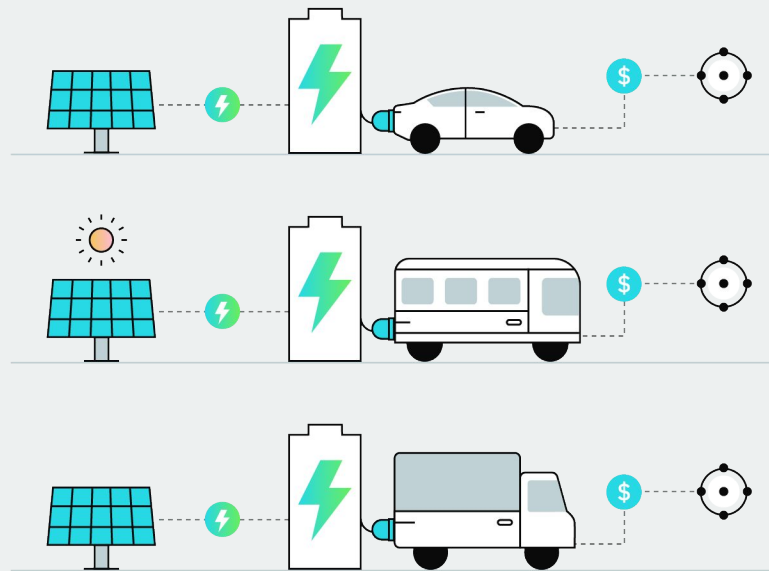
EV TOKEN

Our Customer's strong bet on innovation can lead to next generation power solutions such as EV charging stations and EZ charger for domestic houses.

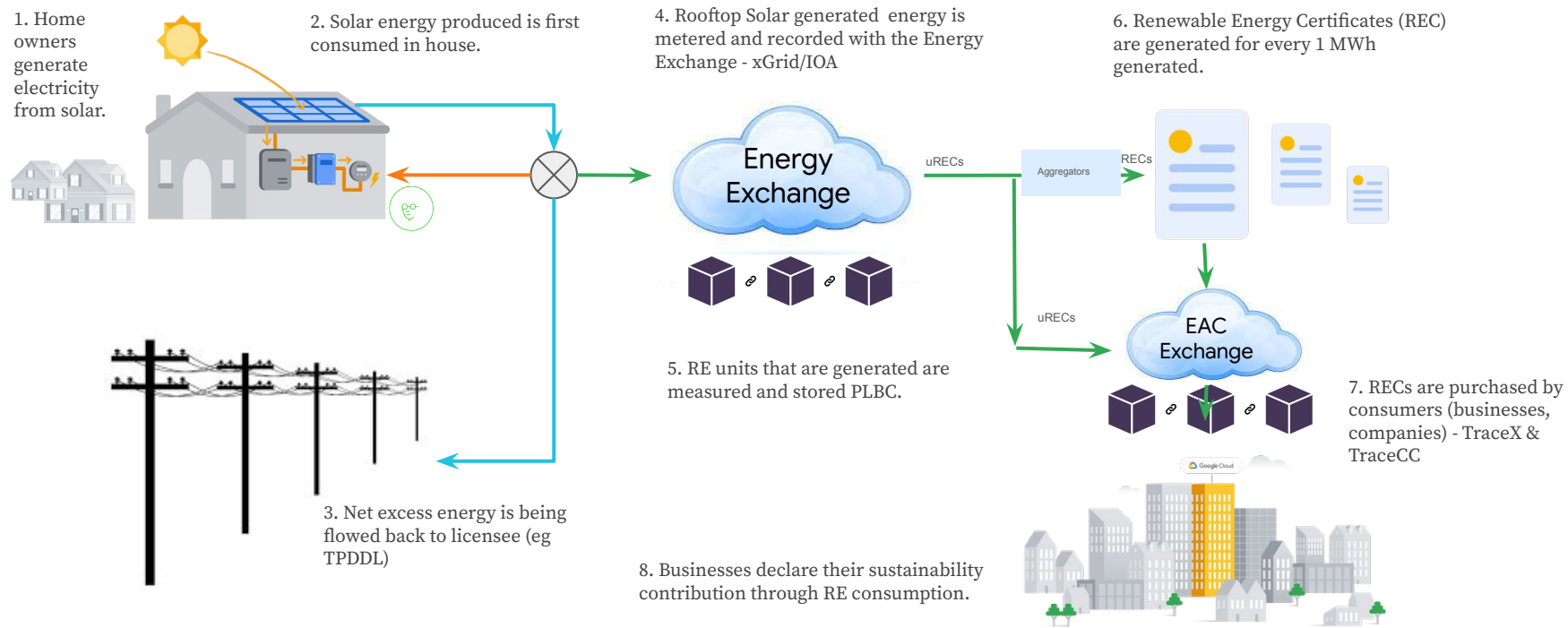
Powerledger has also been strongly involved in EV charging projects in the past and currently works in Blockchain based solutions for optimizing EV charging programs.

EV Charging all along an EV Token provides:

- Ensure **guaranteed green** charging with granularity certification
- Use excess solar from the stores or neighbouring households to provide green charging
- Introduce batteries and extend the utilisation of solar
- Report on the green energy supplied to the EV charging stations
- Create a customer-centered environment for the charging
- Extend the token value to other products such as EZ home
- Seamlessly create audit reports for possible regulatory tax reductions thanks to blockchain technology certification



uRECs aggregation and exchange unlock new value streams and create customer benefits





You have turbocharged your web3 transformation and become a global leader

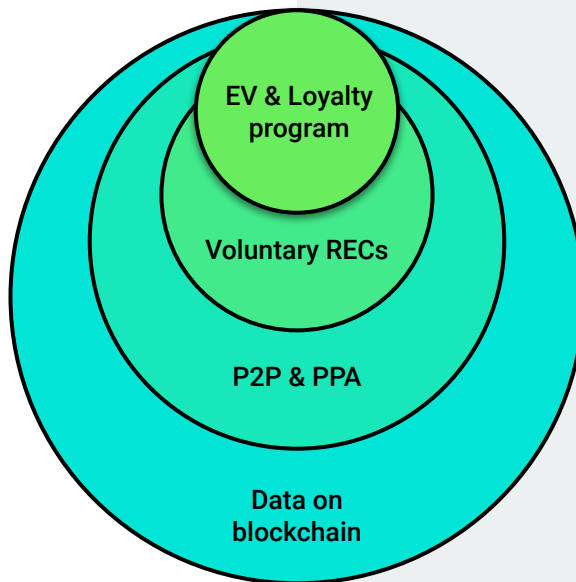
You become a fully web3 based utilities/energy company.

You hold all the data, aggregate all the assets and enable all of the solutions, including marketplace participation.

You become a one-stop shop.

The world will become a lot simpler.

Your customer acquisition strategy can utilise completely different loyalty programs. There is a clear differentiation between you and all the other utilities.



Web3 benefits

- Consumer choice and digital empowerment
- Utilising blockchain for financial settlements and data storage
 - And rewards/loyalty program acquisition
- Unique electricity products that:
 - Connect the existing retail and emerging electricity utilities business models
 - Deliver monetisation or ROI on assets for the customers
 - Are transparent, trustworthy and guaranteed green

Thank you



www.powerledger.io



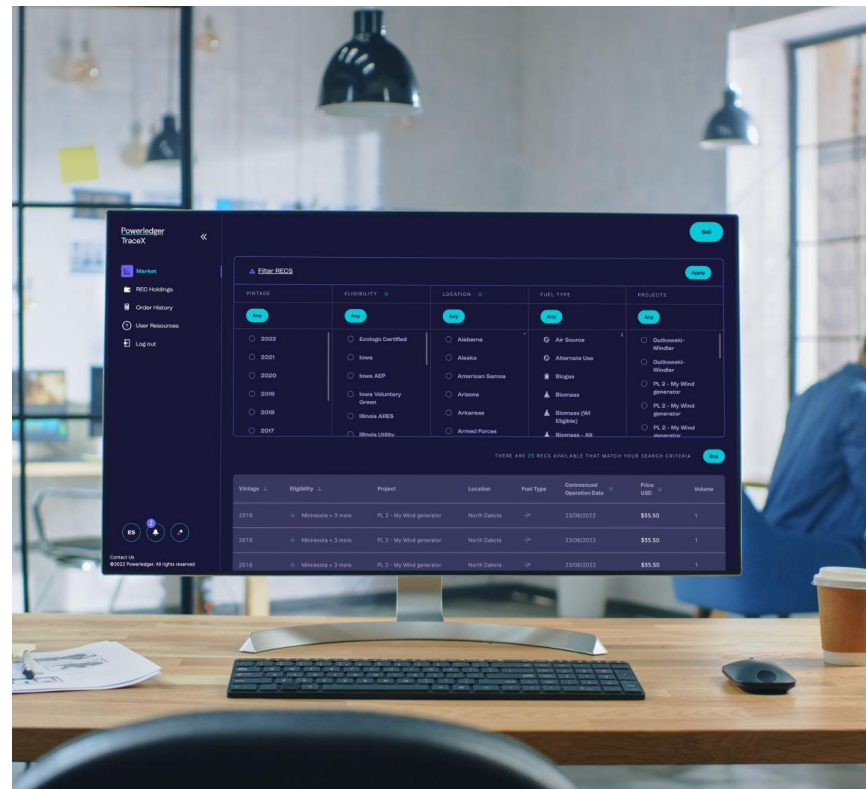
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